

Upscaling Microclimatic Conditions into Body Temperature Distributions of Ectotherms

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ABSTRACT: Realistic projections of the biological impacts of climate change require predicting fitness responses to variations in environmental conditions. For ectotherms, this challenge requires methods to scale-up microclimatic information into actual body temperatures, T_b , while dealing with uncertainty regarding individual behaviors and physiological constraints. Here, we propose an information-theoretical model to derive microhabitat selection and T_b distributions of ectotherm populations from microclimatic data. The model infers the most probable allocation of individuals among the available microenvironments and the associated population-level T_b distribution. Using empirical T_b data of 41 species of desert lizards from three independently evolved systems—Western North America, Kalahari Desert, and Western Australia—we show that the model accurately predicts empirical T_b distributions across the three systems. Moreover, the framework naturally provides a way to quantify the importance of thermoregulation in a thermal environment and thereby a measurement for the constraint imposed by the climatic conditions. By predicting T_b distributions of ectotherm populations even without exhaustive information on the underpinning mechanisms, our approach forms a solid theoretical basis for upscaling microclimatic and physiological information into a population-level fitness trait. This scaling process is a first step to reliably project the biological impacts of climate change to broad temporal and spatial scales.

Keywords: biophysical model, body temperature, ectotherms, behavioral thermoregulation, heat balance, maximum entropy.

Introduction

Predicting the biological impacts of climate change remains a chief priority for ecological research. Reliable projections of these impacts require moving beyond environment-organism correlations and developing methods capable of assessing the effect of climate on fitness-related traits that modulate popu-

lation responses to environmental changes (Kearney et al. 2009; Sinervo et al. 2010; Urban et al. 2016; Nowakowski et al. 2018). For ectothermic organisms, a primary feature linked to endurance to climate is body temperature, T_b , which modulates performance of physiological functions, reproduction, and growth (Angilletta et al. 2002; Sinervo et al. 2010). However, modeling T_b in ectotherms requires integrating fine-scale information on microclimatic conditions with multiple physiological constraints, microhabitat preferences, and behavioral plasticity (Porter et al. 1973; Bakken and Gates 1975; Stevenson 1985; Bakken 1992; Seebacher et al. 1999; Seebacher and Shine 2004; Kearney et al. 2009; Woods et al. 2015; Malishev et al. 2017). Recent advances in biophysical modeling allow one to accurately characterize microclimatic conditions and individual T_b by integrating multiple mechanisms at fine spatial and temporal scales (Kearney and Porter 2017; Malishev et al. 2017). However, in order to generalize predictions to broader ecological scales, these models must balance the complexity of all physiological and behavioral processes that organisms use to adjust T_b (Huey et al. 2009; Kearney et al. 2009; Sinervo et al. 2010) and still make accurate predictions scalable to the population and species levels. Such framework is nonetheless lacking thus far.

At fine scales, animals select microhabitats by balancing multiple costs and benefits on thermal performance, food availability, and risk of predation (Sears et al. 2011; Malishev et al. 2017). For example, animals select foraging patches while avoiding exposure to extreme microclimatic conditions (Sears et al. 2011; Malishev et al. 2017) or optimize the time spent in a refuge by balancing thermal costs and benefits with the risk of predation outside of the refuge (Polo et al. 2005). Moreover, individual decisions are subject to some stochasticity either when there is no preference for one microhabitat or when unexpected events change animal behavior (Woods et al. 2015; Vickers and Schwarzkopf 2016; Malishev et al. 2017). Altogether, those thermoregulatory and nonthermoregulatory decisions determine the daily distribution of body temperatures experienced by an individual or a

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population of ectotherms (Huey et al. 1977; Bauwens et al. 1996; Huey and Pianka 2017). Daily T_b distributions have long been used by field studies with thermoregulating terrestrial ectotherms to infer characteristics of their thermal biology (Huey et al. 1977; Hertz 1992; Hertz et al. 1993; Bauwens et al. 1996; Huey and Pianka 2017), the quality of thermal habitats, or the effectiveness of behavioral thermoregulation (Hertz et al. 1993; Row and Blouin-Demers 2006; Zamora-Camacho et al. 2014). Understanding the processes that modulate animal activity routines, microhabitat selection, and T_b distributions is therefore crucial to improve the predictive capacity of biophysical models.

By integrating multiple mechanisms of microhabitat selection—such as nutritional and heat constraints, costs of displacements, or landscape structure—computational models accurately predict individual activity routines and T_b at local scales (Sears et al. 2016; Vickers and Schwarzkopf 2016; Malishev et al. 2017). However, analyzing behavioral thermoregulation in its full complexity at large spatial and temporal scales rapidly increases the need for prior information from a wide array of systems and species. For example, global-level analyses require knowledge of multiple ectothermic taxa with independently evolved life histories and species-specific thermal constraints. Similarly, increasing the timescale increases the possibilities for unexpected factors or rare events to change model predictions, for example, owing to unexpected climatic variations or habitat modifications (Kearney et al. 2009; Sunday et al. 2014; Nowakowski et al. 2018). Hence, a model capable of making robust predictions on organismal responses to climate variations at broad spatial and temporal scales should integrate such uncertainty involving environmental, physiological, and behavioral processes. To this aim, the maximum entropy framework—a method to infer the most probable density distribution from a small amount of prior information—offers a promising way to make accurate predictions without the need to know the complete repertoire of underpinning mechanisms (Jaynes 1957; see Harte 2011). In ecology, the maximum entropy framework has been applied to understand collective behaviors of animals and thus scale-up individual decisions into population-level descriptors of system functioning (Attanasi et al. 2014).

Here, we present a novel probabilistic model based on the maximum entropy framework to predict T_b distributions of populations of ectotherms that thermoregulate behaviorally in heterogeneous habitats. The model derives the most probable allocation of individuals among the microenvironments available for thermoregulation and estimates their T_b using a transient heat biophysical model. With this information, the model derives the T_b distribution of the population at each instant of time. We validated model predictions using empirical T_b data of desert lizards published by Huey and Pianka (2017). Desert lizards are among the best-

studied examples of thermoregulating ectotherms because they rely on behavior to avoid ambient temperatures beyond their critical thermal limits (Huey et al. 1977; Pianka 1986; Sunday et al. 2014). The data set comprises 171 empirical daytime T_b distributions of 41 species of desert lizards from three independently evolved systems: Western North America, Kalahari Desert, and Western Australia. Our results demonstrate that daytime T_b distributions of the desert lizards match the maximum entropy distribution, that is, the T_b distribution of individuals allocated randomly between the available microenvironments, subject only to the constraint of minimizing the deviation from the preferred temperature (the mean T_b of lizards in the field). Hence, our method allows inferring T_b accurately using only microenvironmental data and preferred temperatures, without requiring further information on physiological, environmental, and behavioral mechanisms. We propose the use of the maximum entropy framework as a robust theoretical approach to scale-up microclimatic and physiological information and project the biological impacts of climate change.

Material and Methods

Modeling T_b Distributions

The aim of this model is to derive the state of a functional variable such as T_b in a population of thermoregulating ectotherms using microclimatic data. Let us consider a large population of ectotherms in which N individuals are allocated among I microenvironments, for example, exposed to the sun, sheltered, or burrowed. We know that T_b of individuals depends on the conditions experienced in each microenvironment, for example, solar irradiation, air, or ground temperature (e.g., Porter et al. 1973; Kearney and Porter 2017). Therefore, there is a T_b distribution that characterizes the entire population at any instant of time, $n(T) \equiv \{T_1, T_2, T_3, \dots, T_N\}$, where T_i are body temperatures of individuals $i = 1, 2, 3, \dots, N$. We can obtain reliable measurements of an average quantity from this distribution, such as the mean T_b of the individuals in the field. A plausible assumption is that mean T_b , $E[n(T)]$, is close to the preferred temperature that individuals target in the field, hereafter preferred temperature T_{pref} (for a discussion of this approach, see appendix, available online). There are infinitely many possible allocation schemes for the N individuals in the repertoire of microenvironments I that are consistent with the thermoregulatory constraint: $E[n(T)] = T_{\text{pref}}$. However, the maximum entropy principle tells us that the most probable allocation scheme is the most stochastic distribution subject to that constraint (appendix; Jaynes 1957; Harte 2011). That is, the T_b distribution that results from allocating individuals randomly among I microenvironments provided the condition $E[n(T)] = T_{\text{pref}}$. Thus, the maximum entropy framework provides an implicit expression for the probabil-

ity of selection of a microenvironment (e.g., sun, shade, or sheltered) within the repertoire I :

$$p(T_i, T_{\text{pref}}, \lambda) = Z^{-1} q_i e^{-\lambda |T_i - T_{\text{pref}}|}, \quad (1)$$

where T_i is the expected body temperature of an individual in the i th environment and Z is the partition function, $Z = \sum_{k=1}^I q_k e^{-\lambda |T_k - T_{\text{pref}}|}$. Probabilities q_i are prior probabilities of an individual in the i th microenvironment and depend on the relative frequency of each microenvironment, for example, the relative availability of shaded or sunlit areas. Parameter λ , which results from applying the method of the Lagrangian multipliers (appendix), relates to both the deviation and the entropy of the probability mass function $p(T_i, T_{\text{pref}}, \lambda)$. For example, when $\lambda = 0$ individuals are equally distributed among the available microenvironments, $p(T_i, T_{\text{pref}}, \lambda = 0) = 1/N$. This means that we have no information about what microenvironment is preferred in terms of thermoregulation; that is, the entropy of the system is maximum. Therefore, the T_b distribution will peak near the mean temperature of the environment, with no influence of T_{pref} (fig. 1). By contrast, increasing λ indicates that animals are more likely to be in the microenvironment that reduces deviations from T_{pref} . In such a case, the T_b distribution will peak closer to T_{pref} and the entropy of the distribution decreases (fig. 1).

The procedure outlined above provides the most probable allocation scheme for N individuals in I microenvironments. However, we still need to derive the T_b distribution of the N individuals at any instant of time. To this aim, T_b distributions were modeled using discrete time integral projection models (Easterling et al. 2000). This procedure allows simulating the dynamics of a continuous trait, such as T_b , in a population knowing the probability of transition between each possible combination of T_b values in the environment. That is, the T_b distribution at any time depends on the distribution at the previous time step multiplied by a probability of transition, or transition kernel (Easterling et al. 2000). This transition kernel is a function of (1) the microclimatic conditions in each possible microenvironment and (2) the probability of selection of each microenvironment, $p(T_i, T_{\text{pref}}, \lambda)$ (eq. 1).

Let $h_k(T_i, T_j, \Delta t)$ be the probability that a transition from a body temperature T_j to T_i takes place in a time step Δt in the microenvironment k , for example, the probability that T_b increases 2°C when an individual remains basking during 1 min. To estimate these probabilities, we need to know the biophysical model, or transient-heat function, that governs T_b variations in each microenvironment, that is, the rate at which T_b approaches the equilibrium T_e . Therefore, the probability of the transition T_j to T_i in the microenvironment k is the product of (1) the probability that such a transition takes place given the biophysical properties of the microenvironment, $h_k(T_i, T_j, \Delta t)$, and (2) the prob-

ability that an animal selects that microenvironment, $p(T_i, T_{\text{pref}}, \lambda)$. Overall, the probability of the transition T_j to T_i in the whole repertoire of microenvironments is the sum of probabilities over the I available microenvironments: $\sum_{k=1}^I h_k(T_i, T_j, \Delta t) p(T_i, T_{\text{pref}}, \lambda)$. This probability gives us the transition kernel that governs the dynamics of the T_b distribution (Easterling et al. 2000). Therefore, the iterative equation for the T_b distribution takes the form

$$n(T_i, t) = \int_{T_{\min}}^{T_{\max}} \left[\sum_{k=1}^I h_k(T_i, T_j, \Delta t) p(T_i, T_{\text{pref}}, \lambda) \right] n(T_j, t-1) dT_j, \quad (2)$$

where $n(T_i, t)$ is the proportion of individuals in the population with body temperature T_i at time t and the integration limits $T_{\min}$ and $T_{\max}$ represent the range of possible body temperatures in the environment. Equation (2) converges to a stable T_b distribution, provided mathematical conditions described elsewhere (Ellner and Rees 2006; see also Caswell 2001). We solved equation (2) numerically using the midpoint rule (Ellner and Rees 2006). This method consists of dividing the temperature range ($T_{\min}$, $T_{\max}$) into m artificial classes and consider the transition kernel as an $m \times m$ matrix, where the entry i, j gives the probability of transition T_j to T_i in the whole environment. We then obtained the stable T_b distribution by iterating matrix multiplications and confirmed model convergence in all the analysis.

Finally, we ask what λ value generates the most probable T_b distribution of a population of thermoregulating terrestrial ectotherms. There are two possible ways to derive this value. (1) Theoretically, on the basis of the maximum entropy principle, the most probable T_b distribution is the most stochastic distribution subject to the thermoregulatory constraint, that is, the one that maximizes entropy and minimizes the deviations from T_{pref} : $\max(S[T_b] - E[|T_b - T_{\text{pref}}|])$ (maximum entropy prediction: λ -MaxS). (2) Empirically, we obtain the λ value that maximizes log likelihood to field T_b data (maximum likelihood value: λ -MaxLik; fig. 2). When correctly implemented, the maximum entropy approach corresponds to maximizing the likelihood that the model generates the observed data (appendix). Therefore, we should find a significant relation between λ -MaxS and λ -MaxLik if the maximum entropy correctly predicts the empirical T_b distribution. Moreover, we analyzed the bias of the observed λ -MaxLik versus predicted λ -MaxS and the goodness of fit of the best-fitted model (i.e., the determination coefficient R^2) of the linear regression between observed and predicted density values.

Body Temperature Data

We used field T_b data of desert lizards recently published by Huey and Pianka (2017) to validate model predictions.

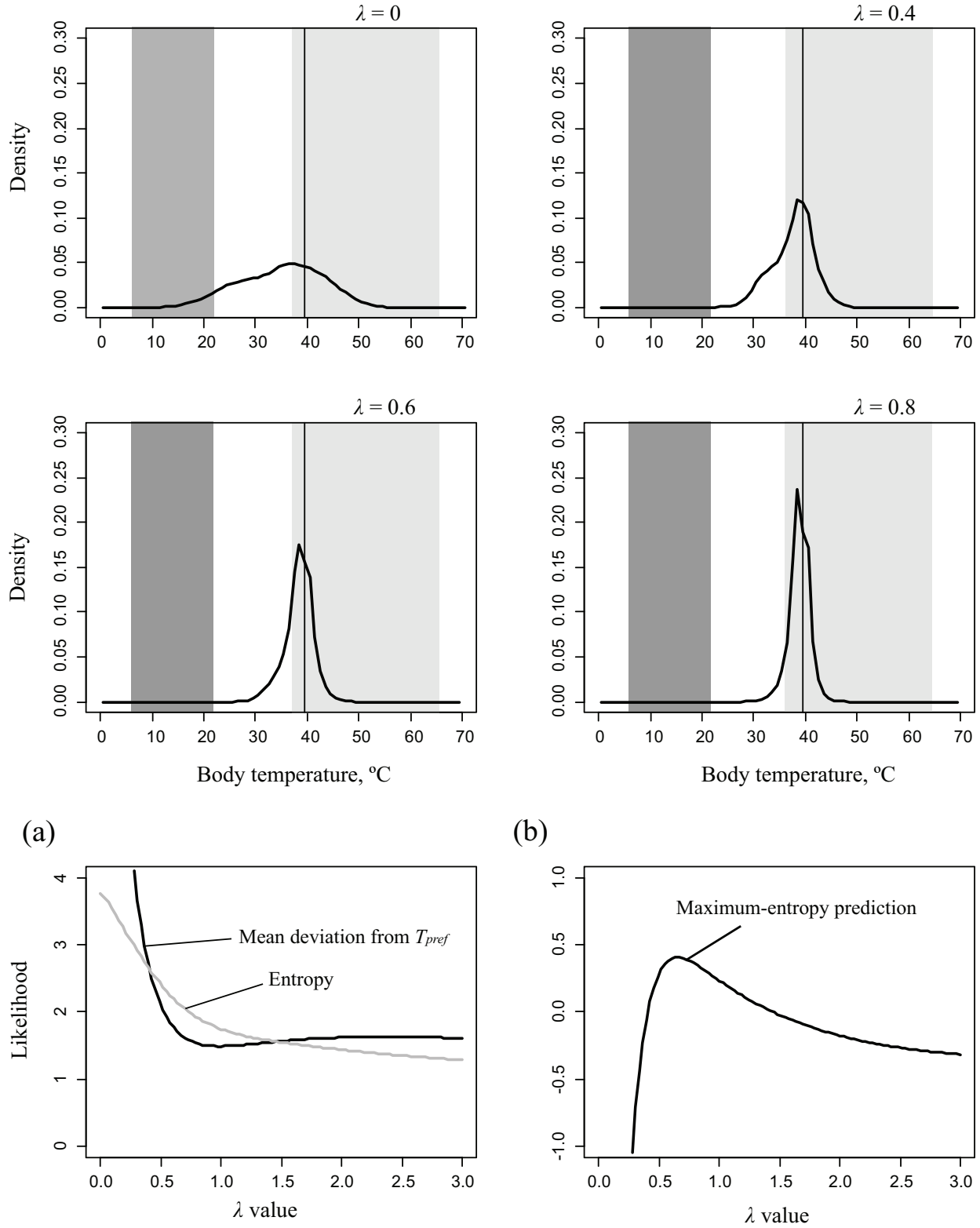


Figure 1: Daytime T_b distributions of *Aspidoscelis tigris* for different values of parameter λ (eq. [1]). Shading represents the daytime variation in the operative temperature of a lizard in the sun (light gray) and in the shade (dark gray); computed using the biophysical model described in “Material and Methods”), and the vertical line is the preferred temperature, T_{pref} . Increasing the λ value reduces both the mean deviation from the preferred temperature, $E[|T_b - T_{pref}|]$, and the entropy, $S[T_b]$, of the distributions (a). The difference $S[T_b] - E[|T_b - T_{pref}|]$ has a maximum at the λ value that maximizes the entropy and minimizes deviation from the optimal temperature (b).

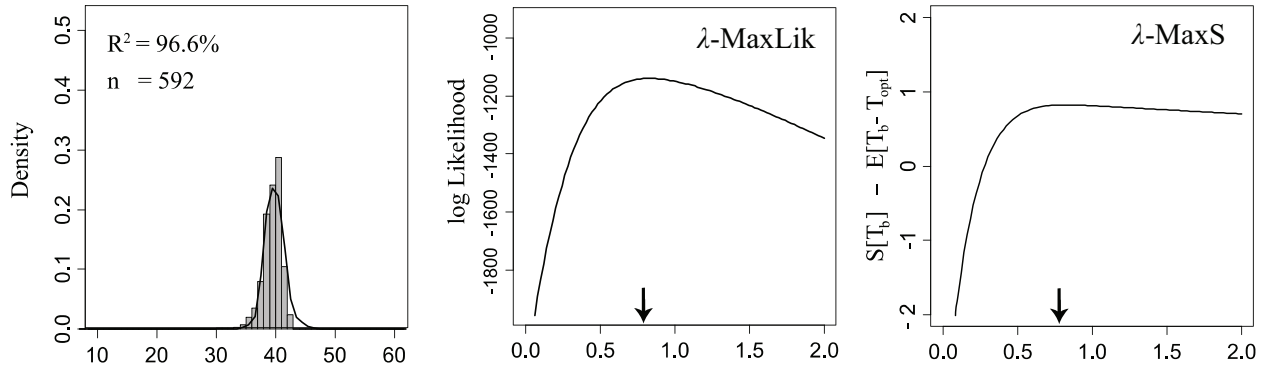
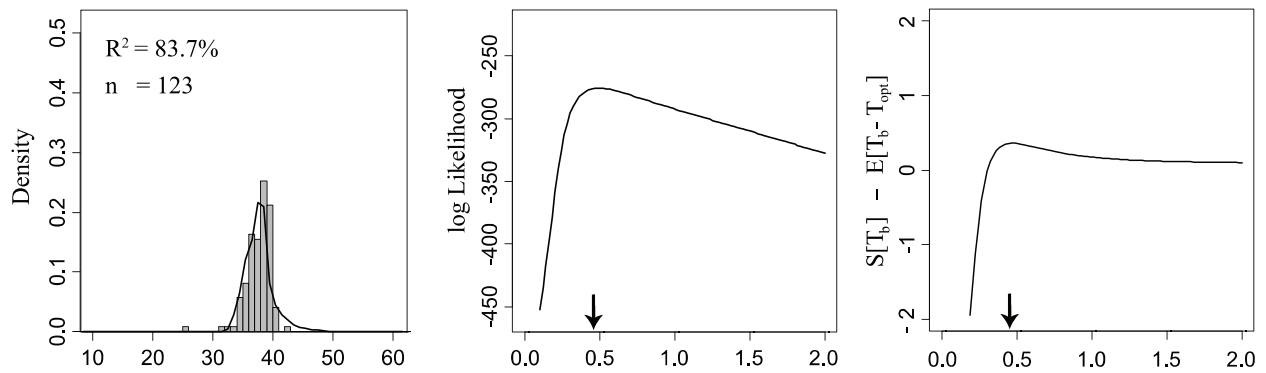
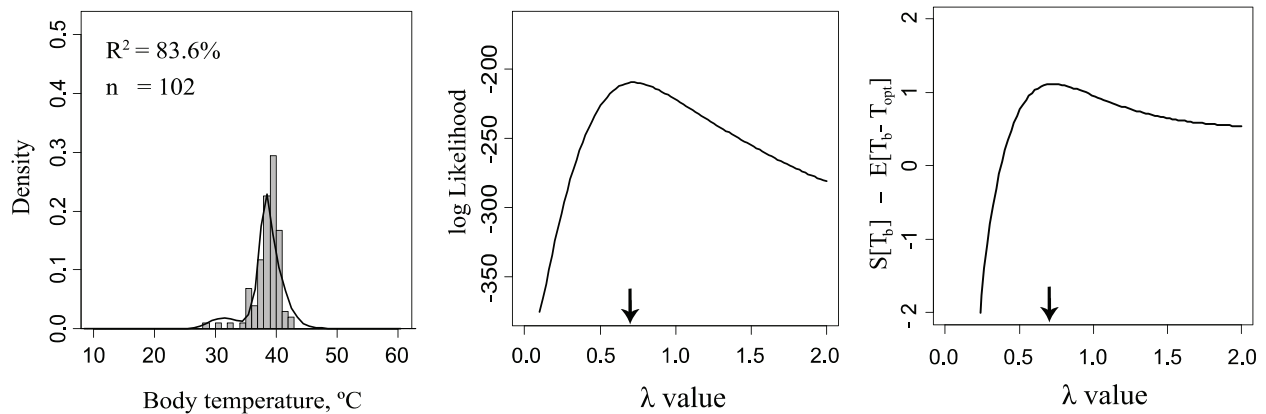
Aspidoscelis tigris (Western North America)*Pedioplanis lineocellata* (Kalahari Desert)*Ctenophorus isolepis* (Western Australia)

Figure 2: Examples of model outputs for each desert (species with the largest sample size; data from Huey and Pianka 2017): T_b distribution (observed histogram and model prediction by maximum likelihood), log likelihood function of parameter λ (arrow indicates λ -MaxLik), and curve of maximum entropy hypothesis (arrow indicates λ -MaxS).

The data set comprises 10 species from Western North America, 10 species from the Kalahari Desert, and 25 species from Western Australia. We built empirical T_b distributions for individual species at each site and within the same month

and considered those with a sample size >10 individuals. In total, we used 171 distributions: 30 from North America (10 species), 61 from the Kalahari (10 species), and 80 from Western Australia (21 species).

We used NicheMapR (Kearney and Porter 2017) to obtain microclimatic information in each location (North America: 42°12'N, 115°50'W to 28°23'N, 111°32'W; Kalahari: 26°S–28°S and 22°15'E–19°50'E; Western Australia: 26°S–28°S and 125°50'E–119°06'E; Huey and Pianka 2017). The package computes hourly microclimate variables using long-term (1960–1991) average monthly values from a global climate database (for details, see Kearney and Porter 2017) and simulates solar radiation, including effects of shading. In particular, we used NicheMapR to compute hourly air and ground surface temperatures, solar radiation, relative humidity, and wind speed at 1 cm in height. We then used a transient heat model parameterized with those microclimatic data to derive T_b in two microenvironments: sun and full shade (i.e., 0% and 100% of shade, respectively). To obtain the relative frequency of sun and shade (probabilities q_i in eq. [1]), we followed the method of Huey et al. (1977). This procedure applies geometrical analyses to calculate the proportion of sunlit areas at various times and seasons as a function of the elevation angle of the sun and plant coverage. We simulated the fraction of sunlit areas of the ground and tree trunk surfaces, respectively, for terrestrial species (31 species) and arboreal species (10 species).

To derive T_b of lizards in each microenvironment, we used a transient heat biophysical model (Porter et al. 1973). The model integrates heat transferred by solar radiation (Q_{sol}), emission of longwave radiation (Q_{IR}), convective heat flow (Q_{conv}), conduction to ground (Q_{cond}), and evaporation of water from the skin (Q_{evap}):

$$\Delta Q = \Delta Q_{sol} + \Delta Q_{IR} + \Delta Q_{conv} + \Delta Q_{cond} + \Delta Q_{evap}. \quad (3)$$

We then derived T_b variations iteratively solving

$$T_{t+\Delta t} = T_t + \frac{1}{cM} \Delta Q \quad (4)$$

(Fei et al. 2012), where c is the specific heat capacity averaged over the tissues and structures of the body and M is body mass (for a description and R script of the transient heat model, see <https://github.com/JRubalcaba/BodyTemp>). The model requires species-specific information on animal body size and morphology. We obtained data on body mass and snout-vent length of each species from Meiri (2010) and used allometric equations to derive the surface area of skin necessary to compute heat transfer. The effective surface area of skin exposed to solar radiation depends on the posture of the animal and its angle with the incident radiation (Porter et al. 1973). Here, we assumed that lizards always expose the entire dorsal surface (two-thirds of the surface area of the skin) to solar radiation and were always in contact with the ground. However, alternative postures and microhabitats

can be integrated into the model by adding multiple combinations of microenvironments and postures to the total repertoire (Rubalcaba et al. 2019).

For the purpose of this model, we need to define T_b variations as a probability $h_k(T_i, T_j, \Delta t)$ of the transition from T_j to T_i in a time step Δt in the microenvironment k (eq. [2]). That is, if the initial temperature is T_b , the most likely temperature after one time step Δt is $\mu(T_{t+\Delta t})$, with some variability σ owing to the uncertainty of the transient heat model. Therefore, if the animal has initial body temperature $T_t = T_j$, the probability of reaching any body temperature T_i is drawn from the Gaussian distribution $N(\mu(T_{t+\Delta t}), \sigma)$, with probability density

$$h_k(T_i, T_j, \Delta t) = \frac{1}{\sigma\sqrt{2\pi}} \exp - \frac{[T_i - \mu(T_{t+\Delta t})]^2}{2\sigma^2}. \quad (5)$$

Fitting the Model

We used the model to generate daily T_b distributions for all species each month for which empirical data are available. We considered T_{pref} to be the mean temperature of the empirical distribution (for alternative approaches, see appendix). With this information, we ran the model to obtain stable hourly distributions and verified model convergence. Then, we integrated hourly distributions over the time interval covered by empirical data (Huey and Pianka 2017) to obtain the daytime distribution. That is, the daytime distribution is a mixture of stable hourly distributions within the period of activity. For each empirical T_b distribution, we built 100 possible theoretical distributions using different λ values within the range (0, 3). We then identified (1) the maximum entropy prediction (λ -MaxS) and (2) the λ value inferred from data (λ -MaxLik). We analyzed the relationship between λ -MaxLik and λ -MaxS using linear mixed effect models including species as a random factor. Finally, we explored the biological significance of parameter λ : the extent to which species need to invest in thermoregulatory activities to control T_b . Because desert lizards rely on thermoregulation to avoid high temperatures, and because this may be especially important in summer (Pianka 1986), we expect λ to take higher values in the summer months (e.g., indicating that they need to spend the central hours of the day away from the sun).

Results

Daytime T_b distributions computed by maximum likelihood (λ -MaxLik) closely matched with empirical data (fig. 2; table 1), and the goodness of fit increased with increasing sample size ($F_{1, 102.1} = 36.16$, $P < .001$; fig. 3). This result demonstrates that parameter λ —which encodes the thermo-

Table 1: Results of models fitted by maximum likelihood: mean λ value of each species in summer and winter, number of distributions used for each species, and mean goodness of fit of models

Species	λ summer	λ winter	n distributions	Mean R^2
Western North America:				
<i>Aspidoscelis tigris</i>	1.514	...	5	88.05
<i>Uta stansburiana</i>	1.230	...	5	82.51
<i>Urosaurus ornatus</i>	1.030	...	1	81.15
<i>Callisaurus draconoides</i>	1.251	...	4	78.85
<i>Dipsosaurus dorsalis</i>	1.678	...	2	62.78
<i>Gambelia wislizenii</i>	1.164	...	4	67.14
<i>Phrynosoma platyrhinos</i>	1.166	...	5	66.64
<i>Sceloporus magister</i>	1.922	...	2	43.07
<i>Uma scoparia</i>	.884	...	1	55.48
<i>Urosaurus graciosus</i>	.743	...	1	53.58
Kalahari Desert:				
<i>Trachylepis sparsa</i>	1.099	1.036	7	70.99
<i>Agama aculeata</i>	1.298	1.125	8	78.55
<i>Meroles squamulosus</i>	1.042	...	2	56.37
<i>Pedioplanis lineocellata</i>	1.130	.683	11	75.58
<i>Meroles suborbitalis</i>	.947	.798	10	71.74
<i>Trachylepis spilogaster</i>	.974	.757	9	59.37
<i>Trachylepis occidentalis</i>	1.402	...	3	63.08
<i>Pedioplanis namaquensis</i>	.707	.666	6	65.15
<i>Nucras tessellata</i>	.844	...	2	64.92
<i>Heliobolus lugubris</i>	1.608	1.000	3	60.82
Western Australia:				
<i>Ctenophorus isolepis</i>	1.015	.742	7	76.07
<i>Varanus eremius</i>	.954	...	1	81.49
<i>Ctenotus leonhardii</i>	1.507	...	3	72.26
<i>Ctenophorus nuchalis</i>	.830	1.829	5	60.12
<i>Varanus gouldii</i>	1.485	1.424	6	63.33
<i>Ctenophorus fordi</i>	...	1.223	2	73.51
<i>Ctenotus grandis</i>	1.062	...	2	57.98
<i>Pogona minor</i>	...	.485	2	71.38
<i>Ctenotus schomburgkii</i>	.800	.498	5	52.62
<i>Ctenophorus scutulatus</i>	1.733	1.160	3	68.32
<i>Lophognathus longirostris</i>	.542	.427	3	63.51
<i>Varanus tristis</i>	...	.904	1	60.30
<i>Diporiphora paraconvergens</i>	...	.281	1	60.18
<i>Ctenotus quatt</i>	1.179	.708	7	73.66
<i>Ctenotus calurus</i>	.984	.763	6	58.11
<i>Ctenotus helenae</i>	1.989	.482	3	75.59
<i>Ctenophorus clayi</i>	1.447	.814	2	55.05
<i>Ctenotus brooksi</i>	...	.668	2	58.84
<i>Ctenotus dux</i>	.922	1.065	7	59.67
<i>Ctenotus pantherinus</i>	.556	.515	4	47.55
<i>Moloch horridus</i>	1.438	1.236	8	42.66

regulatory constraint of ectotherms in their environment—can be inferred from field T_b data using maximum likelihood methods. Moreover, empirical distributions fitted with maximum likelihood (λ -MaxLik) were significantly related to the theoretically derived maximum entropy distributions (λ -MaxS; $F_{1,31.4} = 603.9$, $P < .001$; fig. 3). The bias of the λ -MaxS predictions with respect to the distributions inferred from data, $|\lambda\text{-MaxLik} - \lambda\text{-MaxS}|$, decreased with increas-

ing sample size ($F_{1,110.4} = 4.28$, $P = .04$). Therefore, even in the absence of T_b data, the maximum entropy framework provides a means to derive T_b distributions of ectotherms using information on the microclimatic conditions of their environment (fig. 3).

The values of the thermoregulatory constraint (parameter λ of the λ -MaxLik distributions) were significantly higher in summer than in the winter months in the Kalahari

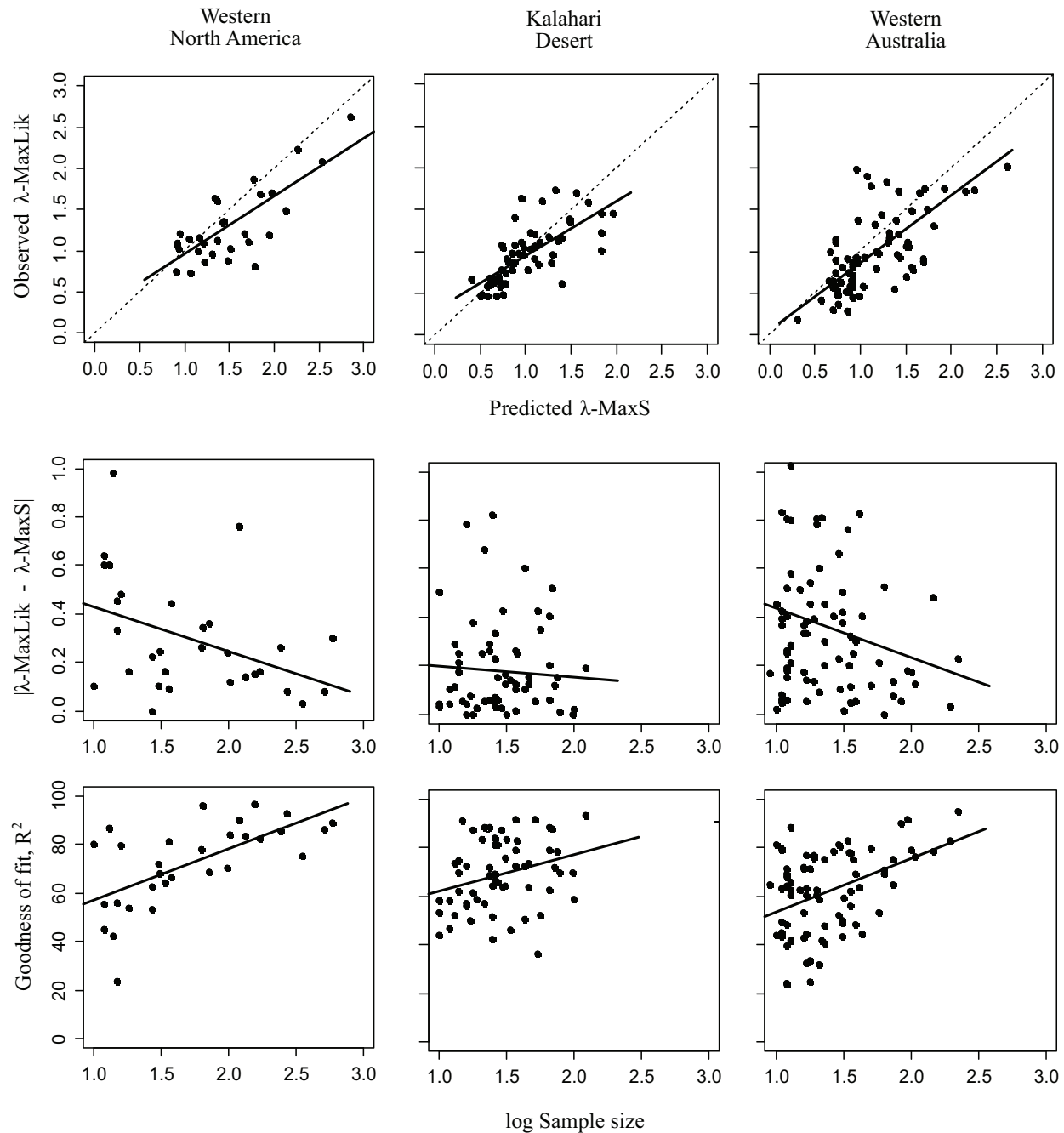


Figure 3: *Top*, relationship between the empirically and theoretically derived models (λ -MaxLik vs. λ -MaxS) for each desert. *Middle, bottom*, deviation from predicted λ value and goodness of fit of the λ -MaxLik model in relation to sample size in the original data (data from Huey and Pianka 2017).

Desert and Western Australia (1.26 ± 0.075 vs. 0.95 ± 0.078 ; $F_{1,149.4} = 12.58$, $P < .001$; table 1; no winter data are available for North American sites; Huey and Pianka 2017). The value of λ -MaxLik did not depend on sample size ($F_{1,165.9} = 0.32$, $P = .57$); thus, this result is consistent irrespective of the goodness of fit of the maximum likelihood model.

Discussion

Biophysical models have provided important advances toward a mechanistic understanding of thermoregulation in ectotherms (Porter et al. 1973; Shine and Kearney 2001; Kearney et al. 2009; Sears et al. 2016; Malishev et al. 2017).

Having a high level of realism, these models suffer from the need for a high number of parameters to be directly measured and require a large amount of species-specific information. Here, we propose an alternative information theoretical approach to infer T_b distributions of thermoregulating ectotherms using microclimatic information from their thermal environment. Our results suggest that behavioral thermoregulation follows a maximum entropy process, in which movements among microenvironments are influenced by thermoregulatory constraints. The strength of these constraints can be quantified through a single parameter λ , which measures the importance of behavioral thermoregulation for an ectotherm in its environment. This model introduces a novel conceptual framework to understand and simulate behavioral thermoregulation in ectotherms and provides a way to simulate the response of ectotherms to contemporary as well as future climates.

There is an increasing demand to obtain information on the quality of thermal habitats and the effectiveness of behavioral thermoregulation to buffer ectotherms against climate change (Huey 1991; Hertz et al. 1993; Huey et al. 2009; Kearney et al. 2009; Sunday et al. 2014). This information is crucial because differences in behavioral plasticity reflect the actual sensitivity of species to the surrounding, heterogeneous thermal environment. However, in order to extrapolate biological information from observed T_b distributions, we need to compare the observed T_b data with appropriate null models, such as inanimate objects (Heath 1964), or random walk simulations (Woods et al. 2015; Vickers and Schwarzkopf 2016). Using the maximum entropy framework, a solution to this issue emerges naturally through the parameter λ (appendix). As λ encodes the constraint for the maximum entropy distribution, it measures the extent to which the T_b distribution differs from the theoretical null distribution of a similar ectotherm moving randomly across the same thermal habitat. Thus, higher λ values indicate that selection of microenvironments is more strongly influenced by thermoregulatory than by nonthermoregulatory behaviors. Consistent with this idea, we observed that λ takes higher values in the summer months (table 1); that is, T_b distributions in summer are further away from a theoretical null distribution. Although this result suggests that thermoregulatory constraints are higher in summer, it does not discard the fact that desert lizards also thermoregulate actively in winter, spending time in the sun to increase body temperature (Huey et al. 1977). Because λ can be estimated from data, it provides a powerful and general measurement of thermoregulatory constraints that is supported by the well-established theoretical background of maximum entropy (Jaynes 1957).

Daytime T_b distributions of diurnal lizards are generally unimodal, left-skewed curves (Huey and Pianka 2017). This shape is important because the T_b distribution determines

the integrated performance or fitness, that is, the average thermal performance that individuals experience during the daytime (Martin and Huey 2008; Vasseur et al. 2014; Buckley et al. 2015). The causes and fitness consequences of the skewness have long attracted the attention of thermal biologists (Heath 1964; Martin and Huey 2008; Vasseur et al. 2014; Huey and Pianka 2017). Among other hypotheses, it has been argued that the skewness results from the daytime variation in ambient temperatures (Heath 1964), behavioral thermoregulation, or the asymmetry of thermal performance curves (for a review, see Huey and Pianka 2017). Most daytime distributions predicted by our model exhibit some degree of skewness owing to the low modal temperatures of hourly distributions in the morning. Interestingly, the degree of skewness is influenced by the thermoregulatory constraint value (parameter λ), which suggests that behavioral thermoregulation plays a role in determining the skewness (Heath 1964; Huey and Pianka 2017; see also appendix). The shape of the T_b distributions differs between species within deserts owing to differences in body mass, shape, or color or differences in microhabitat use and thermoregulatory behaviors (Huey et al. 1977; Huey and Pianka 2017). Although we considered differences in body mass and shape, preferred body temperature, and microhabitat (see "Material and Methods"), we made simplifications that may have partially obscured the interspecific variability. Thus, there are deviations from observed distributions that cannot be simply explained by differences in sample size (e.g., distributions with the same sample size are less biased in the Kalahari than in Australia or North America; fig. 3). Although the maximum entropy method can derive T_b distributions with little species-specific information, providing a more detailed biophysical model or a better description of the structure of microenvironments and the repertoire of behaviors can improve the accuracy of its predictions. Ultimately, increasing the quality of the prior information may allow us to better analyze the biological components of the T_b distributions.

The effectiveness of behavioral thermoregulation depends on the fine-scale structure of the thermal landscape and the availability of microenvironments for thermoregulation. Empirical studies and models demonstrated that landscape structure influences T_b distributions of terrestrial ectotherms and determines their potential to regulate T_b behaviorally (Sears 2005; Sears et al. 2016). Landscape structure determines the availability or frequency of microenvironments in the repertoire and the way these frequencies vary with time of day and season (Huey et al. 1977). Individual-based models that simulate organisms moving across their thermal landscapes have successfully implemented landscape structure by integrating spatially explicit information (e.g., Woods et al. 2015; Sears et al. 2016). In our model, this information is a prior probability distribution that describes the frequency of each

microenvironment in the repertoire (appendix). Although we used a geometrical approximation to derive the relative frequency of sunlit and shade areas in the ground and tree trunks (following Huey et al. 1977), the model can also integrate spatially explicit data to parameterize the structure of microenvironments when such information is available. In this sense, our model provides a way to understand the influence of spatial structure on behavioral thermoregulation using information theory. Thus, for an animal moving randomly across microenvironments, the thermoregulatory constraint is uninformative ($\lambda = 0$). Therefore, the shape of its T_b distribution may be fully determined by landscape structure (the prior probability) and the thermal inertia of the body, approaching an average distribution of operative temperatures in its thermal landscape (Hertz et al. 1993; Woods et al. 2015). By contrast, increasing the thermoregulatory constraint ($\lambda > 0$) reduces the importance of the prior information on landscape structure because animals may actively seek microenvironments preferred for thermoregulation. The maximum entropy method provides a framework to further investigate the strength of the thermoregulatory constraints of an ectotherm in its thermal environment and the sensitivity of the T_b distribution to variations in landscape structure.

In summary, using the maximum entropy framework, we offer a probabilistic model to infer T_b distributions of ectotherm populations using microclimatic data. Predicting the response of ectotherms to climate change requires methods to handle multiple physiological and behavioral mechanisms and stochastic processes that interact in a complex way. However, gathering such detailed data for several species across multiple habitats and periods is technically prohibitive. In this sense, our approach offers a solid basis to infer the most probable T_b distributions by maximizing the entropy subject to the thermoregulatory constraint. This framework benefits from recent methodological advances on biophysical modeling that allow simulating microclimatic conditions (Kearney and Porter 2017) and fine-scale thermal refuges from climatic data (Milling et al. 2018). Furthermore, published databases offer information on preferred temperatures, field T_b , or critical thermal limits of many ectothermic species (Sinervo et al. 2010; Huey and Pianka 2017; Bennett et al. 2018), which can provide further empirical support to our framework. Most importantly, predicting actual T_b of ectotherm populations, our method makes it possible to reliably project the biological impacts of climate change on these organisms by linking biophysical, physiological, demographic, and climatic information.

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